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|---|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   |   | into $V_s$ of water. Hence when the ice cube totally melts, the water level will remain the same.                                                                                 |
| 7 | A | <p>By conservation of energy,</p> <p>Initial potential energy = final kinetic energy + work done against friction</p> $(1.0)(9.81)(1.0) = KE + (0.5)(10.0)$ $KE = 4.81 \text{ N}$ |
| 8 | B | <p>Let the extension of the spring when it is at equilibrium be <math>x</math></p> <p>Loss in GPE = <math>mgx</math></p>                                                          |